GUARDRAILS

What VITALS will never do.

Make up a source. The model returns a condition id; the link is looked up from the knowledge base.

Name a condition it wasn't given. Inventions are dropped and reported.

Show a percentage. Confidence is a word: strong match, possible, unlikely.

Fake a vital sign. No sensor means “not connected.”

Stay quiet when the backup check worries. If the rule engine escalates, the answer escalates.

TEAM TETHER

PHOTO

Joshua Collado
Lead · reasoning engine

PHOTO

Joaquin
Medical research · rules

PHOTO

Cruz
UI design · user testing

Ms. Sylvia Odoms
Sponsor · [SCHOOL NAME]

Scan for the website

blackphenixlord.github.io/vitals-site

[TEAM EMAIL]

A student project for NASA HUNCH. Decision support, not a diagnosis, and not a medical device.

NASA HUNCH 2026–27

When the doctor is twenty minutes away.

Offline medical decision support for long-duration crews.

TEAM TETHER

Software & Technology

THE PROBLEM

Far from Earth, help takes time a crew may not have.

3–22 min

One-way signal delay between Earth and Mars.

Offline

No internet. It all runs on one small computer on board.

Plain words

Crew describe symptoms the way people really talk. VITALS asks one follow-up at a time.

HOW IT WORKS

An AI that has to show its sources.

01 **Listen.** Findings are read from the complaint and shown back, so a misreading can be fixed.

02 **Narrow.** 14 spaceflight conditions and 72 findings, each tied to NASA sources.

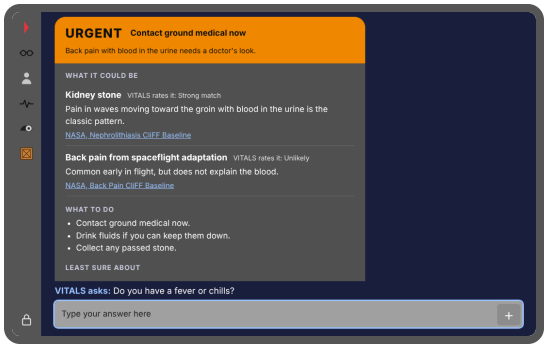
03 **Reason.** A local model ranks only those conditions.

04 **Double-check.** A rule engine always votes on how serious it is.

05 **Answer.** A level, what to do now, and a source link for every condition.

!! EMERGENCY | URGENT | MONITOR | ROUTINE

THE ASTRONAUT SCREEN



Demo case: flight day 63, pain in waves, blood in the urine. Big tap targets, high contrast, works offline.

MEASURED · 26 SEP 2026

Right condition, 8B model	**22 of 26**
Right condition, 3B model	**19 of 26**
Missed escalations	**0**
Automated tests	**175**

Next: teach it to say “nothing fits,” run it on the Jetson, add sensors, get clinician review.